

EpidemIQs: Prompt-to-Paper LLM Agents for Epidemic Modeling and Analysis

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Abstract—Large Language Models (LLMs) offer new opportunities to accelerate complex interdisciplinary research domains through AI assistant tools. Epidemic modeling, characterized by its complexity and reliance on network science, dynamical systems, epidemiology, and stochastic simulations, represents a prime candidate for leveraging LLM-driven automation. We introduce EpidemIQs, a novel multi-agent LLM framework that integrates user inputs and autonomously conducts literature review, analytical derivation, network modeling, mechanistic modeling, stochastic simulations, data visualization and analysis, and finally documentation of findings in a structured manuscript, through five predefined research phases. We introduce two types of agents: a scientist agent for planning, coordination, reflection, and generation of final results, and a task-expert agent to focus exclusively on one specific duty serving as a tool to the scientist agent. The framework consistently generated complete reports in scientific article format. Specifically, using GPT 4.1 and GPT 4.1 Mini as backbone LLMs for scientist and task-expert agents, respectively, the autonomous process completes with average total token usage 870K at a cost of about \$1.57 per study, successfully executing all phases and final report. We evaluate EpidemIQs across several different epidemic scenarios, measuring computational cost, workflow reliability, task success rate, and LLM-as-Judge and human expert reviews to estimate the overall quality and technical correctness of the generated results. Through our experiments, the framework consistently addresses evaluation scenarios with an average task success rate of 79%. We compare EpidemIQs to an iterative single-agent LLM, benefiting from the same system prompts and tools, iteratively planning, invoking tools, and revising outputs until task completion. The qualitative and quantitative comparisons suggest a consistently higher performance of the proposed framework across the tested scenarios. We discuss the benefits that EpidemIQs can offer, its limitations, and ethical concerns of such AI research assistant tools.

Impact Statement—EpidemIQs advances autonomous assistant tools through an innovative and transparent orchestration of collaborative multi-agent LLMs and integration of multimodal data. To the best of the authors' knowledge, this framework is among the first in the field of epidemic modeling that leverages AI for an end-to-end autonomous workflow. Our evaluations demonstrate the framework's high task success rate, bringing us

closer to efficient, reliable, and accessible copilots for complex interdisciplinary research.

Index Terms—Autonomous Agents, AI for Science, Epidemic Modeling, Multi-Agent Systems

I. INTRODUCTION

Scientists' efforts to automate research date back to at least the early 1970s [1]–[3], aiming to overcome limits on the number of ideas scientists can pursue, constraints that force them to prioritize ideas with higher predicted impact [4], [5]. For example, the Automated Mathematician was introduced to carry out simple mathematics research guided by large sets of heuristic rules [4], [5], and DENDRAL was proposed to assist chemists with data interpretation problems [6].

More recently, advances in autoregressive large language models (LLMs) [7]–[9] have demonstrated strong capabilities in question answering, problem-solving, and coding, with promising applications across science and engineering. However, LLMs still face limitations when applied to real-world tasks [10]. These constraints led to the development of LLM agents, which extend LLMs with abilities such as external tool use [11], [12], chain-of-thought prompting [13], and iterative self-improvement [14]. LLM agents have since been applied to diverse domains, including software engineering [15], [16], cybersecurity [17], medical diagnosis [18], chemistry [19], [20], material science [21]–[23], computational biology [24], algorithm design [25], and healthcare [26]. Multi-agent systems have also emerged to simulate human-like research environments across disciplines. Examples include Agent Laboratory [27], Virtual Lab [28], ChemCrow [29], ResearchAgent [30], and The AI Scientist [31]. For broader surveys of LLM agent applications, see [32].

Despite the stunning advances of LLM agents across various domains, epidemiology-related areas have not yet fully benefited from these developments [33], largely because they are among the most sophisticated and interdisciplinary fields. Network-based epidemic modeling is among the sophisticated approaches to studying the spread of infectious diseases by incorporating the structure of contact networks, which represent interactions among individuals or groups in a population [34].

Unlike traditional compartmental models, which assume homogeneous mixing, network-based models account for heterogeneous contact patterns by representing individuals as nodes and their interactions as edges (possibly weighted or time-varying) in a graph. This approach captures the realistic

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social and spatial structures influencing disease transmission, such as clustering, degree distribution, and community structures. By integrating network topology, these models provide insights into how connectivity patterns affect epidemic dynamics, enabling more accurate predictions of disease spread, the identification of critical transmission pathways, and the evaluation of targeted intervention strategies, such as vaccination or social distancing, in complex populations [34]. Addressing these models requires an interdisciplinary expertise: (i) stochastic-process theory to analyze the master equations and their approximations; (ii) network science to characterize and parametrize contact structures; (iii) epidemiology to constrain pathogen-specific parameters; and (iv) computational epidemiology to conduct simulations and scenario analysis. Having a research group with this level of diversity is not only challenging to access but also poses coordination challenges. In this study, we introduce *EpidemIQs*, a novel multi-LLM agents system that can execute an end-to-end workflow to address network-based epidemic modeling with minimal human intervention.

EpidemIQs emulates an interdisciplinary research environment with five collaborative teams of *task-expert* and *scientist* agents, capable of performing end-to-end workflow autonomously: given a user query, if identified as a network-based problem, it initiates the process by discovering relevant insights from diverse online sources (e.g., web-based content and published papers) to conduct a literature review and address analytical aspects through mathematical reasoning. It then formulates the problem as a network-based model, tests it through rigorous stochastic simulations, analyzes the resulting multimodal data (e.g., visual and numerical), and finally composes a scientifically formatted manuscript that reports the entire procedure, from problem formulation to simulation results, discussion, and final conclusion. The framework can also operate in copilot mode, which enables human cooperation throughout the process.

To demonstrate the capability, we test our proposed framework across ten scenarios, focusing on five main aspects of epidemic scenarios articulated through questions, three of which were unknown to the framework. Other scenarios, including real-world outbreak data, opinion dynamics, and financial contagion, that share similar dynamics are presented in the Supplementary Material. We tested *EpidemIQs* against the iterative single-agent as a baseline for ablation of orchestration, comparing performance by (i) discussing the results of each team across all scenarios, (ii) AI and human expert evaluations of generated results and papers, (iii) workflow completion and task success rates, and (iv) computational statistics. The proposed framework consistently showed impressive performance, achieving a 100% workflow completion rate and $79 \pm 8.6\%$ task success rate, an average human review score of 7.98 ± 0.2 out of 10, a low cost of only \$1.57, and a total turnaround time of 1,190 seconds. We can summarize the main contributions of our work as follows:

- Developed novel multi-LLM agents that conduct epidemic

research through five distinct scientific phases.

- Integrated multimodal data (visual, textual, numerical, graph structures, etc.) from diverse sources, including online retrieval, experimental results, and internally generated artifacts (e.g., graphs, tables), to enhance utility and adaptability across various research contexts.
- Enabled fully autonomous mode to minimize human intervention for workflow completion, requiring only an initial query, while also supporting a copilot mode for collaborative operation.
- Ensured high interpretability by having each scientist agent provide reasoning for its actions and generated outcomes, and recording results and scripts to facilitate validation of procedures and outcomes.

The remainder of this paper is organized as follows: Section II describes the methodology. Section III details the design and implementation of the system. Section IV outlines the experimental setup, evaluation cases, and metrics. Section V presents the ablation study, results, and discusses the study's limitations and shortcomings. Finally, Section VII concludes the paper and proposes directions for future work.

II. METHODOLOGY

LLMs demonstrate advanced capabilities relevant to scientific tasks, including coding and answering technical questions. However, their performance in tackling complex epidemiology scenarios remains limited. To address these challenges, we propose a multi-agent framework that emulates a scientific epidemic research laboratory environment to enhance the practical capabilities of LLMs for complex scientific tasks. The *EpidemIQs* framework is organized into four core functional layers that collectively support autonomous workflow [32]: (i) The multi-agent orchestration layer. (ii) The backbone LLM. (iii) The perception layer. (iv) The action layer executes tasks derived from the LLM's output, encompassing activities such as code generation, simulation, modeling, etc.

Figure 1 illustrates how these layers interact to support an end-to-end workflow. The orchestration layer assigns tasks and synchronizes agent interactions so that different subtasks are handled coherently. The perception layer converts diverse inputs into unified semantic representations that can be reasoned over. Based on these inputs, the backbone LLM analyzes the evidence, forms hypotheses, and determines the sequence of actions required to address the research question. The action layer then autonomously calls appropriate tools, such as stochastic simulators, Retrieval-Augmented Generation (RAG), API calls, and executes them in the correct order.

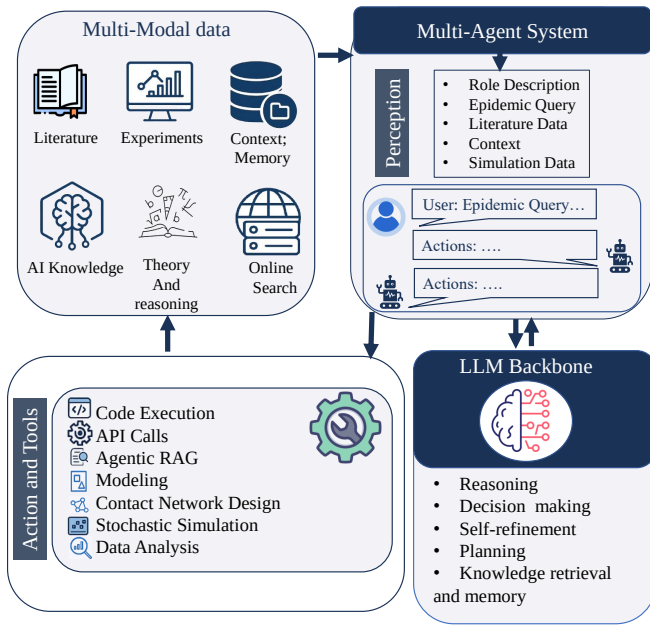


Fig. 1: Multimodal approach for the multi-agent framework for flexible epidemic research. This framework enhances the power of multi-LLM agents by integrating multimodal data from various sources, such as literature, online web data, theoretical models, and simulation and experiments.

Through this process, the system translates a high level research question into executable steps and produces results that directly address the posed objective.

III. MULTI-LLM AGENT ARCHITECTURE: EPIDEMIQS FRAMEWORK

The orchestration of EpidemIQs is structured to emulate a scientific laboratory, comprising five phases: Discovery, Modeling, Simulation, Analysis, and Report Writing. These phases are executed by distinct collaborative multi-agent system teams, and the results of their work are integrated to produce the final report. In this section, we first describe the design of two types of agents, followed by the architecture design, output structure, memory, and implementation details of EpidemIQs.

A. Two Agent Types: Scientists and Experts

Two agent types are inspired by the generative agent concept [35] and further advanced by frameworks such as CAMEL [36], AutoGen [37], and Metagpt [38], and are designed to enable autonomous scientific reasoning and investigation. The central coordinating component, referred to as the *scientist* agent, performs high-level orchestration through iterative planning, reflection, and refinement of execution, as shown in Figure 2. Upon receiving an input query, the scientist agent employs a dedicated *plan* module, which parses and decomposes the query into a plan comprising sub-tasks and their logical dependencies. Both *reflect* and *ReAct* modules are constrained to generate output in predefined formats. While structured outputs, e.g., JSON, enhance reproducibility and reliability of performance, they have recently been shown to compromise the reasoning capability of LLMs [39]. Therefore, we explicitly devote a

separate plan module as an agent to allow it to freely generate the plan, while both the subsequent *ReAct* and *reflect* modules are restricted to generating predefined structured outputs. Once

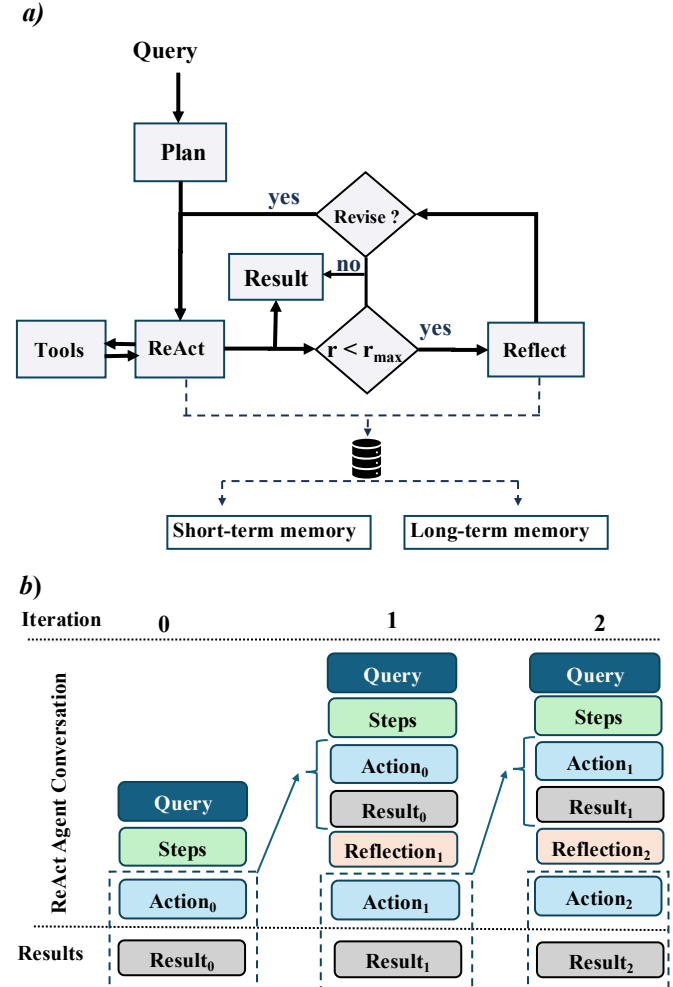


Fig. 2: The scientist agent architecture. a) Three modules as plan, ReAct, and reflect. The reflection ends either when the reflect agent decides the answer is already good enough or it reaches its maximum iteration limit. b) the conversation components for ReAct agent, as can be seen each time with the fixed task and plan, it updates new actions, results, and reflection. At each iteration t , the newly generated $Action_t$, corresponding $Result_t$, and $Reflection_t$ are appended to short-term memory, while the $Action_{t-2}$, corresponding $Result_{t-2}$ are dropped out. The loop continues updating the conversation trace until the aforementioned conditions are met.

the plan is formulated, the scientist agent iteratively executes the ReAct-Reflect loop until either the reflect agent decides no further revision is needed, or it reaches the maximum iteration limit, r_{max} . The ReAct Module executes reasoning, and call tools conditioned on the current plan segment, and refine its response based on the action outcomes, while the reflect module prompted to perform logical reasoning, evaluates intermediate outputs for errors, logical inconsistencies, or

incomplete results, producing structured JSON feedback to refine action selection and output validity across iterations, enhancing the agent performance through linguistic feedback without need to update the weights of models. In parallel, the scientist agent can coordinate multiple independent *task-expert* agents as tools, each designed as a specialized ReAct agent for a specific task or data source. These expert agents serve as domain specialists responsible for well-defined atomic tasks such as literature retrieval, online search, mathematical derivation, or data extraction. This functional decomposition ensures that each expert agent executes its task with minimal context switching, while the scientist agent maintains global task coherence and quality control.

B. Multi-Agent Orchestration

Figure 3 illustrates that EpidemIQ's framework operates after receiving the user query, beginning with *Discovery*. Here, the *Discovery Scientist* utilizes a Multi-Hop Question Answering (MHQA) paradigm [40] across three iterative steps to progressively refine data collection accuracy [41]. The team comprises:

- *Discovery Scientist*: Coordinates expert agents to synthesize a structured output defining the scenario, task specifications, and disease context.
- *Online Retriever Expert*: Fetches and refines relevant data from web sources via external APIs (e.g., Tavily).
- *Literature Retriever Expert*: Aggregates and summarizes pertinent academic findings via the Semantic Scholar API.
- *Math Expert*: Provides quantitative analysis and Python-based execution to enhance accuracy [42], utilizing reasoning capable LLMs (e.g., OpenAI-o3-Mini).

Subsequently, the *Modeling* phase constructs the components for simulating epidemic dynamics through three specialized agents:

- *Network Scientist*: Models the population contact network (e.g., random, scale-free, or multilayer [43]) using scientific libraries like NetworkX.
- *Model Scientist*: Defines the mechanistic disease progression framework by selecting appropriate compartmental models based on Discovery phase insights.
- *Parameter Scientist*: Calculates and assigns quantitative model parameters and initial network states via Python code execution.

The framework then transitions to *Simulation*, executing stochastic experiments via the Fast Generalized Epidemic Modeling Framework (FastGEMF) [44]. The *Simulation Scientist* configures the engine for multi-compartment processes on multilayer networks, recording time-series data and state-evolution visualizations. The Vision Expert verifies simulation outcomes to ensure validity. In the *Analysis* phase, a multi-agent team derives actionable insights:

- *Data Scientist*: Coordinates the team to compute epidemiological metrics (e.g., peak infection size, outbreak probability) and synthesize qualitative analysis.
- *Vision Expert*: Interprets visual outputs (e.g., trajectory uncertainty bands) to validate dynamic behaviors not easily deduced from raw numbers.

- *Data Expert*: Analyzes structured numerical data stored in the repository using Python libraries (Pandas, SciPy) via a multi-hop strategy.

Finally, *Automated Report Generation* synthesizes findings into an academic manuscript. The *Reporter Expert* drafts sections (e.g., Methods, Results) using a LaTeX template, while the *Latex Craft Expert* validates syntax and structure. The Literature Review Expert integrates related work for contextual positioning. The draft undergoes an iterative self-correction loop (n_{max} cycles) before final PDF compilation. The details of the architecture are provided in Supplementary Section S1.1.

C. Memory

Two specialized memory structures are considered for the agents. The scientist agent's memory consists of short-term and long-term memory. The short-term memory encompasses all the current conversation details, including role description, input query, agent's internally generated responses, interaction with tools, conversations with other agents, and the final output. The long-term memory stores all previous conversation history in a database, enabling memory retrieval based on the current query.

D. Structured Outputs and Operation Modes

Typically, LLMs generate free-format texts as a natural language output, which does not follow a strict structure. However, the performance of the collaborative LLM agents requires precise and careful interoperability, as information sharing and communication can significantly impact overall outcomes [37]. Therefore, to make the data transition between each phase of the process robust, reliable, and more predictable, there are constraints defined for each team output structure, forcing them to produce output in a predefined structure (e.g., JSON schema). This not only facilitates data retrieval and improves prompt development efficiency, but it also enables automatic validation of each team's output [45], ensuring all required fields are generated accordingly. If not, the agent retries until it fulfills the task or reaches the maximum number of tries allowed. While the output structure of each scientist agent contains different fields, they all share a common element: the reasoning and logical justification behind their choices and results. Each scientist agent is expected to defend their decisions against hypothesized criticisms, demonstrating why these were the most appropriate outcomes.

Furthermore, EpidemIQs can operate in two modes: end-to-end autonomous and cooperative. The latter is named as a copilot, which needs the user to be actively in the loop to proceed. The autonomous mode does not require human intervention during pipeline execution; instead, it initially provides the query. The copilot mode enables the human to intervene, providing feedback and instruction on the agent's output or asking for additional actions to guide the agent toward the desired direction. After each phase, the human is asked to review the results, and they can either mark them as complete and proceed to the next phase or ask for new actions by providing comments.

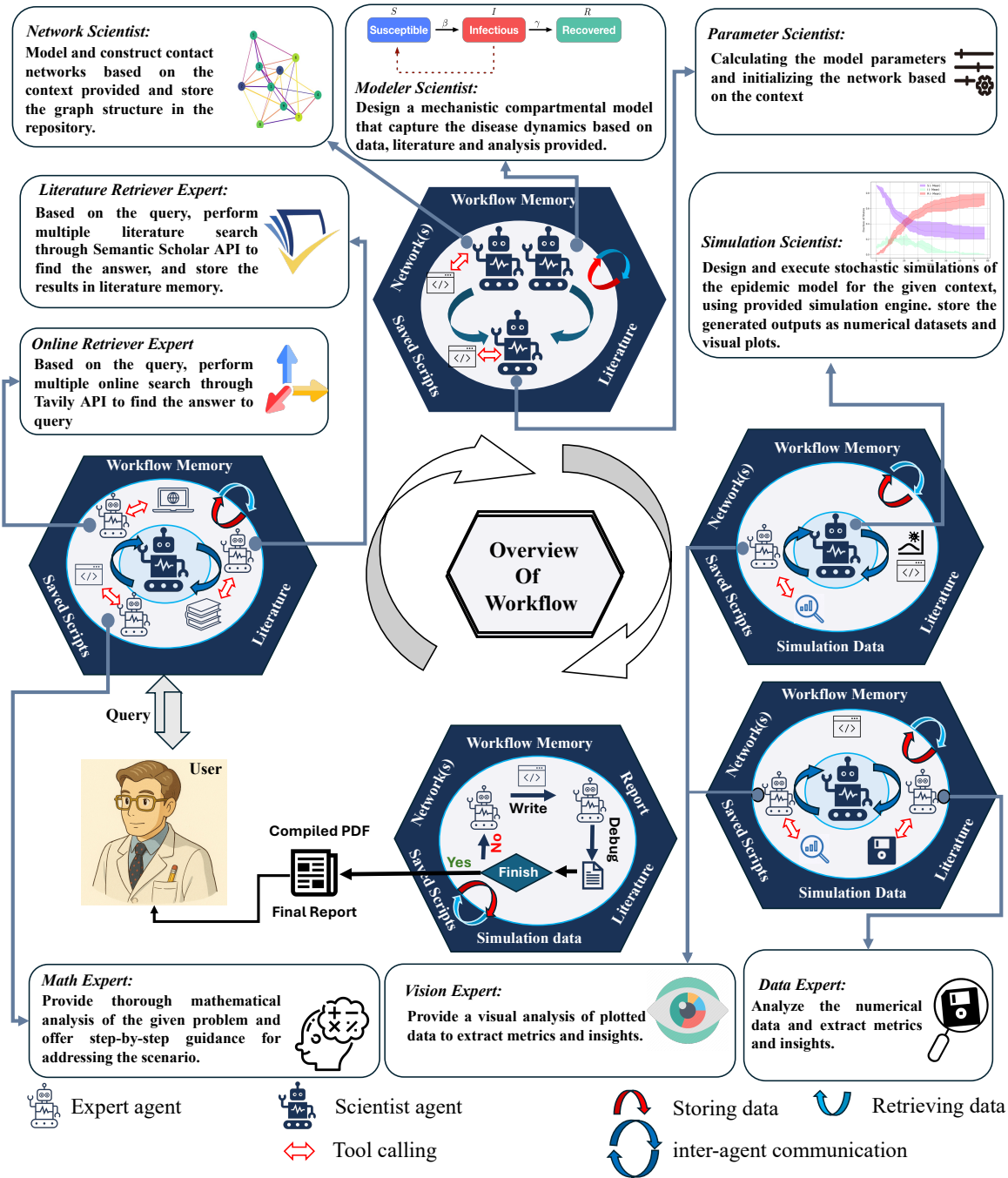


Fig. 3: EpidemIQs orchestration across five research phases. Specialized agent teams perform tool calling (via red arrows), collaborate with each other (curved blue arrows), and exchange artifacts through a shared project repository (hexagons), enabling an end-to-end pipeline from discovery to final report generation.

E. Implementation

The framework is implemented in Python by the Pydantic AI framework. Autonomous data acquisition is supported through API integrations using the Semantic Scholar API for literature retrieval and the Tavily API for web-scale information access. Vision-Language Model (VLM) is implemented via the OpenAI SDK with BinaryContent format to enable efficient visual data interpretation for multimodal processing. Short-term memory utilizes the LLM's context window, while long-term memory is stored in JSON and managed through Pydantic

AI memory functions. Finally, we used Pydantic data models for structured outputs for schema validation. For complete implementation details, including pinned software versions, random seed configurations, and a manifest of all experimental artifacts, refer to Supplementary Section S1.2.

IV. EVALUATION APPROACH

To demonstrate how the framework addresses various tasks, we adopted five epidemic questions (Figure 4), with increasing depth and complexity, to cover different aspects

of epidemic modeling. The first question in Figure 4 is the

Question 1

What is the effect of incorporating degree-heterogeneous networks structure in an SEIR model on disease dynamics, compared to a homogeneous-mixing network?

Question 2

What is the reason for The chain of transmission to break? the decline in infectives, or a complete lack of susceptible?

Question 3

In an activity-driven temporal network with 1000 nodes, where each node activates with a rate of $\alpha=0.1$ and forms transient connections upon activation, how does the temporal structure of the network influence the spread of an infectious disease modeled using the SIR model with a basic reproduction number $R_0 = 3$, compared to its corresponding time-aggregated static network in which edge weights represent the frequency of interactions over time?

Question 4

Competitive SIS model over two multiplex layer A and B same set of nodes. the two viruses are exclusive: A node cannot be infected by virus 1 and virus 2 simultaneously with rates β_1 and δ_1 for layer 1 and β_2 and δ_2 for layer 2. Assume the effective infection rates of each virus are larger than their no-spreading mean field threshold, i.e., $\tau_1 > 1/\lambda_1(A)$ and $\tau_2 > 1/\lambda_1(B)$. Will both viruses survive (coexistence), or will one virus completely remove the other (absolute dominance)? which characteristics of a multilayer network structure allow for coexistence?

Question 5

An epidemic has a reproductive number of $R_0 = 4$, i.e. the mean excess degree of the effective network across which it is spreading is $q = 4$. Let's say the mean degree of this network is $z = 3$ and there are no degree correlations. You want to stop the epidemic by vaccinating people, using a vaccine that gives sterilizing immunity, so that a vaccinated person cannot transmit the disease. What proportion of the nodes do you need to vaccinate if you vaccinate random people? Also what proportion of nodes would you need to vaccinate if you are just vaccinating people who have a degree of exactly $k = 10$?

Fig. 4: Evaluation questions designed to assess the capabilities and limitations of the proposed framework across varying levels of complexity

simplest. It requires the framework to analytically investigate the individual behavior effect on spread, design simulation setup by creating the SEIR model and at least two contact network structures, one heterogeneous and one homogeneous, to compare the stochastic results. The goal is to analyze how structural heterogeneity in host contact patterns shapes population level disease dynamics.

The second focuses on identifying why the chain of transmission breaks and validating hypotheses through stochastic modeling [43]. The third scenario introduces a temporal contact pattern, which exceeds the scope of the provided simulation tools and instruction, testing system adaptability [46]

The fourth demands both analytical reasoning (or knowledge from literature [47]), an extension of the SIS model to a competitive SI_1I_2S form on a multiplex network, exploring conditions for coexistence, and simulation on multilayer networks, for which no instructions were provided regarding the multilayer networks setup. The fifth is the most challenging, requiring analytical evaluation of random versus targeted vaccination strategies. Here, the key challenge is to notice the implicit constraint to ensure that the contact network contains enough degree-10 nodes to make targeted vaccination feasible while preserving its structural properties (January 2025 question of the month from Network Science Society). Collectively, these questions span the key aspects of network epidemics, such as topology, temporal structure, multilayer interactions and competing pathogens, termination mechanisms,

network-aware interventions, physics-aware contact network, etc, providing a holistic evaluation framework. We emphasize that these scenarios function as architectural stress tests for the autonomous research pipeline; they are intended to demonstrate the framework's methodological coherence, distinct from the empirical calibration required for real-world epidemiological forecasting. Only the first two questions were involved in the test and trials through the design of the EpidemIQs orchestration, and the last three were unknown to the framework and were out of the immediate scope of the tools or system prompts provided to the system.

Having a standardized benchmark for evaluating an agentic AI system for science remains an open and non-trivial challenge. Unlike coding, retrieval, or mathematical tasks with binary success conditions, scientific research is inherently open-ended. Recent works have adopted varying assessment strategies. For example, focusing on the overall usefulness of the results [27], [48], relying exclusively on AI-based judgment [31], labeling randomly sampled statements from reports as correct or incorrect [49], or highlighting successful outcomes in specific case studies to demonstrate applicability [23], [28], [29]. In epidemiology, this challenge is compounded by the stochastic nature of the problem and the existence of multiple valid modeling approaches for a single query. Therefore, there is no single ground truth against which to measure performance mechanically. Consequently, we adopted a multi-faceted assessment strategy aligned with the scientific peer-review process, evaluating both technical validity and report quality. We considered the following metrics:

- **Human Expert Evaluation** to measure overall correctness and quality. We employed blinded peer review by domain experts to assess the generated manuscripts based on the criteria in Supplementary Table S10. They also inspected the generated artifacts (code, plots, and simulation data) to verify *Technical Soundness* and *Experimental Rigor*, ensuring the reported results were supported by the underlying work and that they could obtain consistent results through the same random seeding of executed experiments. They also evaluated the quality of the reports and their relevance and alignment with results.
- **LLM-as-Judge Evaluation** using GPT-4o to autonomously evaluate the generated papers using the same assessment criteria as human reviewers, providing a scalable metric for relative quality comparison.
- **Workflow Completion Rate** is defined as the proportion of trials in which EpidemIQs autonomously executed the full workflow and produced a scientific report containing all pre-specified sections, independent of content quality.
- **Task Success Rate**, where success is defined as a binary metric evaluated by human experts based on two criteria: (i) the scientific correctness of the conclusion, and (ii) whether the conclusion is substantiated by the stochastic simulation results. This metric provides a feasible approach to analyze

the approximate usefulness and correctness across all trials.

- **Computational Costs** by tracking the turnaround time, total token usage, and API costs to assess the efficiency of the framework.

While these measures provide comparative performance indicators (descriptive quantitative statistics and rubric-based judgments) rather than statistically validated benchmarks, they capture both the scientific quality of the outputs and the reliability of the autonomous workflow. For baseline configuration to evaluate the architectural benefits of the EpidemIQs orchestration, we compared our framework against a robust, multi-turn single-agent baseline. To ensure a fair comparison, this baseline was designed with structural guidance equivalent to EpidemIQs. The single-agent operates iteratively, autonomously planning its next steps up to a maximum limit, s_{max} . It is provided with the same curated workflow instructions, phase-specific structured prompts, and definitions of required output fields as the specialized agents in EpidemIQs. However, to strictly avoid penalizing the baseline, we modified the prompts to remove multi-agent specific role designations (e.g., You are a full professor in ...), as these shifting persona constraints within a single context window would risk inducing role confusion or behavioral inconsistency. Instead, the single-agent acts as a unified scientific entity, focusing purely on the procedural instructions and objectives of each phase.

Furthermore, the single-agent is equipped with the identical toolset used in the multi-agent framework. This includes access to Vision Expert, which offloads the token-heavy process of image analysis. To ensure the agent's capability to handle the load of handling all scientific phases within a single context, we used LLMs with large context capabilities: OpenAI gpt-4.1-2025-04-14 (≈ 1 million token context window with reasoning capabilities) and o3-2025-04-14 (200K token context window and high reasoning capability). To mitigate context drift as a common issue in long-horizon tasks, we implemented a memory reinforcement mechanism. In each turn, the primary objective is re-injected into the context as ****Reminder**** main query is {query}. Do NOT lose sight of this. This setup ensures that the baseline represents a strong comparison between monolithic context coherence versus the distributed attention mechanism of the EpidemIQs multi-agent architecture. We provide a comparison between their performance based on computational statistics, performance analysis, and review scores of the generated results.

The setup used for assessment is presented in Table I.

TABLE I: Configuration Details for EpidemIQs

Parameter	Scientists	Experts	Math Expert
LLM	gpt-4.1-2025-04-14	gpt-4.1-Mini-2025-04-14	o3-Mini-2025-01-31
r_{max}	1	-	-
Retries	5	5	5
Tool Retries	50	50	50
Tool timeout (sec.)	360	360	360
Output Type	Pydantic dataclass	free-form	free-form

V. RESULTS AND DISCUSSION

Across the five epidemic scenarios, EpidemIQs demonstrated high architectural stability, with our evaluation indicating a workflow completion rate of 100% across 100 trials. Automated rubric scoring by the LLM-as-Judge assigned a mean quality score of 9.04 ± 0.21 out of 10, while four network science Ph.D. students and one faculty reviewer gave an average expert rating of 7.98 ± 0.35 , praising the reports' methodological soundness, clarity, and depth. The task success rate varied based on the complexity of the task and the alignment of the provided tools. For the first two scenarios (Questions 1 and 2), the framework successfully addressed them in approximately 90% of trials. The temporal contact pattern (Question 3) presented the greater challenge, with a task success rate of 65%. For identifying viral coexistence regions (Question 4), it achieved 75%, and for calculating targeted vaccination thresholds (Question 5), 80% of the time it could calculate and validate vaccination thresholds, resulting in an average of $79 \pm 8.60\%$ overall task success rate. In this section, first we briefly discuss the successful results of the framework, followed by an ablation study V-B, limitations V-D, a detailed failure taxonomy V-C1, and finally, ethical concerns are discussed V-E).

Question 1. EpidemIQs successfully validated the fundamental impact of network topology on disease spread dynamics. By modeling two contrasting topologies, a heterogeneous scale-free network and a homogeneous complete graph, the framework demonstrated how degree heterogeneity and superspreader nodes significantly alter epidemic dynamics compared to well-mixed models. The stochastic experiments (300 realizations) and the results analysis confirmed that structural heterogeneity alters the final epidemic size, peak infection size and time, etc., aligning with theoretical expectations. Figure 5 illustrates the Network Scientist's reasoning, highlighting how the agent justified its modeling choices to capture realistic contact

NetworkScientist(GPT-4.1) Reasoning for Scenario 1

"Design logic: - The complete graph exactly models homogeneous mixing, ensuring all disease transmission is equally probable between any two nodes (matches mean-field ODE assumptions; perfect for theory and as a simulation baseline).

- The scale-free configuration model network closely follows a power-law with exponent 2 and a mean degree near the desired value, resulting in high heterogeneity that is typical of empirical human contact networks, ensuring realistic and insightful contrasts for epidemic outcomes.

- Degree sequences were generated carefully (inverse sampling + tail cutoffs + mean tuning) for realism and to enforce graphical constraints. Multiedges/self-loops were removed for population realism.

- Degree moments and distributions were verified for accuracy, and histograms confirm theoretical network structure (homogeneity vs heterogeneity).

- Both networks are ready for direct input into SEIR simulation and analytical comparison protocols specified in the experiment with full control over network effects on epidemic dynamics.

- All major code, metrics, and outputs were saved for reproducibility, enabling later review or improvement."

Fig. 5: An example of reasoning information Network Scientist generated for Question 1. This highlights how the agent justifies the selection of scale-free networks to model heterogeneity against a homogeneous baseline.

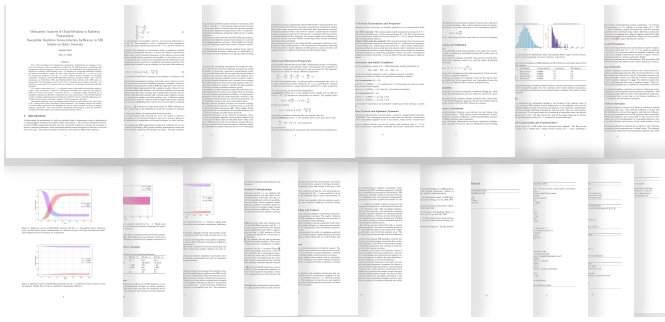


Fig. 6: An example of paper generated by EpidemIQs for the second question.

patterns. Full workflow details and the generated report are available in Supplementary Section S1.4.

Question 2. The framework correctly determined that transmission chains break when the effective reproduction number drops below unity ($R_e < 1$), rather than solely due to the complete depletion of susceptibles. To validate this, the system engaged an SIR model across both supercritical ($R_0 = 3$) and subcritical ($R_0 = 0.5$) regimes on Erdős-Rényi and Barabási-Albert networks. The resulting manuscript (visualized in Figure 6) presented a robust analysis showing that outbreaks can self-extinguish due to low transmissibility or network constraints even when a large fraction of the population remains susceptible. This demonstrates the framework's capacity for validating counterintuitive epidemiological concepts through rigorous simulation, as discussed in detail in Supplementary Section S1.5

Question 3. This scenario tested the system's adaptability by introducing a temporal contact network that is out of the scope of the provided simulation tools. EpidemIQs successfully derived a mean-field approximation for the epidemic threshold. It identified that the temporal ordering of contacts imposes structural constraints that increase this threshold. By comparing the temporal network against a time-aggregated static counterpart (where edge weights represented interaction frequency), the analysis revealed a clear divergence: the temporal network resulted in smaller and slower outbreaks. Critically, the framework demonstrated awareness of its computational constraints, explicitly noting in the final report that the provided engine (FastGEMF) was restricted to static experiments, necessitating a custom simulation algorithm and comparative analytical approach, as detailed in Supplementary Section S1.6

Figure 7 displays the results and process pipeline for Question 4. In this scenario, the framework successfully identified the criteria for the coexistence of competitive exclusive viruses over multilayer networks and extended the analytical mean-field approach in the literature. It demonstrated the dominance, coexistence, and extinction regimes in stochastic processes by designing contact network properties that enable coexistence and varying the model parameters. Full detailed results are provided in Supplementary Section S1.7. Finally, for Question 5, the framework not only determined the vaccination thresholds for random and targeted vaccination strategies (75% and 11.25%, respectively), but also carefully



Fig. 7: An overview of the workflow for Question 4, highlighting each step's goals, main actions, and outputs. The results have been truncated for clearer visualization. At the bottom, plots generated during the modeling and simulation phases are provided, showing extinction, coexistence, and dominance regions.

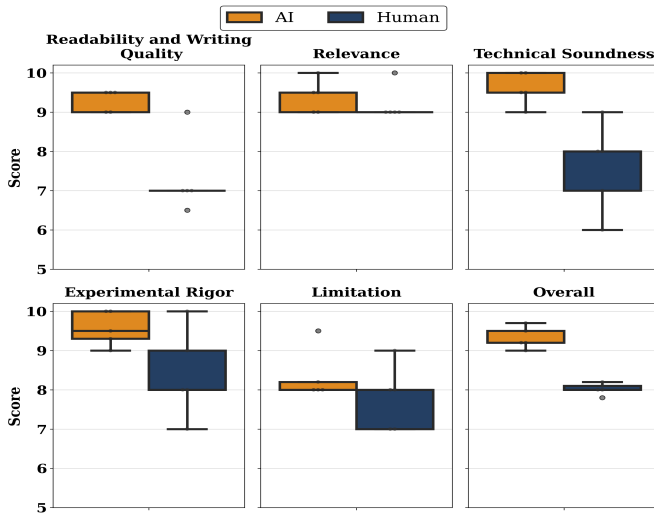


Fig. 8: Evaluation results of five epidemic questions by AI and human reviewers across different rubrics.

modeled a configuration network to preserve the aforementioned centrality measures while ensuring a sufficient number of nodes to verify the targeted vaccination threshold. Stochastic realizations demonstrated the threshold's effectiveness by examining outcomes in regions both above and below it. Supplementary Figure S13 presents the truncated workflow results for Question 5, with additional details provided in Supplementary Section S1.8.

A. Review Score and Computational Cost

Figure 8 presents the evaluation scores assigned by an LLM-as-Judge and by human experts across five rubric criteria. For each of the five epidemic questions, the framework autonomously executed the pipeline 20 times, yielding 100 project repositories, including reports and artifacts. We randomly selected five of them for blind human review. Each reviewer scored every criterion on a 1-to-10 scale. Human scores averaged 7.98 ± 0.20 , indicating consistently strong performance by EpidemIQs. Reviewers praised the comprehensive problem framing in manuscripts, robust experimental design and analysis, and balanced coverage of both analytical and stochastic aspects. The main criticisms concerned verbosity with occasional repetition of content, the use of uncommon vocabulary that slowed reading, and some missing or incomplete figure references.

Table II reports the average turnaround time, total number

TABLE II: Average Per-Phase and Total Computation Statistics

Phase	Time(sec.)	Tokens(K)	Cost(\$)
Discovery	158	98	0.1
Modeling	140	175	0.72
Simulation	198	88	0.23
Analysis	177	93	0.16
Report Writing	517	416	0.36
Overall	1190	870	1.57

of tokens, and total cost (calculated by OpenAI's May 2025 pricing) for all questions, measured from the initial query to

the final report. A total cost of only \$1.57, a turnaround of under 30 minutes, a 100% workflow completion rate, and high review scores demonstrate the strength of the framework and the efficiency of its multi-agent orchestration. The low cost and rapid execution are achieved by splitting tasks between expert and scientist agents. Token-heavy, low-complexity jobs are delegated to fast, inexpensive models such as GPT-4.1 Mini, while more demanding tasks such as planning, reasoning, tool coordination, and output refinement are handled by the full GPT-4.1 model. To approximate the human-equivalent effort for EpidemIQs' average performance, we adopt a similar approach from task duration baselines from Model Evaluation and Threat Research (METR) [50]. Our estimate is approximately 38.4 hours for a complete execution cycle, from literature retrieval to final reporting. A detailed breakdown is provided in Supplementary Section S1.3

B. Ablation Study

To better understand what drives the EpidemIQs' performance, we conducted an ablation study isolating the contribution of each scientist module. Next, evaluating the iterative single-agent performance offers a meaningful baseline for better understanding the added value of EpidemIQs orchestration mechanism.

1) *Scientist Agent Modules Analysis*: To systematically analyze how different parts of the scientist agent contribute to performance, we conducted an ablation study on the DSBench on data analysis tasks [51]. DSBench is a recent benchmark for advanced data science tasks that requires multimodal inputs such as figures, text, and tabular data. In all configurations, we keep the backbone model fixed to OpenAI GPT 4.1 Mini, and we only change the internal architecture of the scientist agent. We first consider a simple LLM-only setting, where the model receives the question and the data but no explicit tool use or control over its own reasoning. In this case, the average accuracy over the 48 DSBench challenges is 29.76%. Adding the ReAct style tool use and intermediate reasoning increases accuracy to 37.40%, which shows that even a relatively light agent structure already helps the model use the data in a more systematic way. Next, we enable the planning module. This configuration, with planning and ReAct but zero reflection step, reaches 44.49%. When reflections were introduced with one step but without planning (ReAct/Reflect, reflection steps=1), performance reached 46.60%. Combining all modules with one reflection step (Plan/ReAct/Reflect, reflection steps=1) achieved 47.70%. Finally, we increase the reflection depth to five steps in the full configuration with planning, ReAct, and reflection. This full scientist architecture achieves 51.71% accuracy, which is the best result across. The gain from one to five reflection steps is smaller than the earlier jumps, suggesting that the benefits of additional reflection begin to plateau, though they continue to yield improvements in accuracy. Figure 9 summarizes the benchmark results. Overall, results show a mostly monotonic improvement when we move from a plain LLM-only agent to a structured scientist agent with tools, planning, and deeper reflection, and it confirms that all three modules are helpful in practice for non-trivial data analysis

tasks, with 73% improvement of accuracy compared to using only plain LLM (from 29.76% to 51.71%).

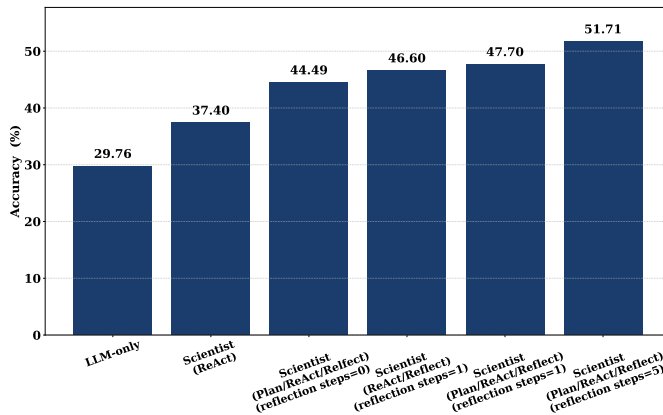


Fig. 9: Accuracy of different configurations of scientist agent over 48 DSbench data science challenges, using OpenAI GPT 4.1-Mini as backbone LLM. The systematic increase in accuracy (from 29.76% to 51.71%) demonstrates that incorporating planning and reflection modules significantly enhances the agent's performance compared to a standalone LLM.

2) *Iterative Single-Agent vs EpidemIQs Orchestration*: We ran the iterative single-agent across the same five questions ten times, using two OpenAI LLMs, GPT-4.1 and o3. Single-agent-GPT-4.1 achieved an average workflow completion rate of $78 \pm 7.7\%$ and an average human-review score of five sampled successful results of 5.06 ± 0.32 , whereas single-agent-o3 reached $80.0 \pm 6.32\%$ with a human-review score of 5.68 ± 0.37 . The *LLM-as-Judge* provides comparative evaluations of the successful papers generated by single-agent and EpidemIQs. Table III presents these scores for all questions. The task success rate was $22 \pm 6\%$ and $26 \pm 9.75\%$ for GPT 4.1 and o3, respectively, as backbone LLMs. Specifically, in none of the trials could they achieve the correct targeted vaccination threshold in Question 5, and only about 10% of the time the stochastic coexistence regime in Question 4 was correctly validated.

Human reviewers observed large performance gaps across the three implementations, even though the AI-generated scores were close. In the single-agent GPT-4.1 setting, the framework could partially address the first three questions, but it often handled the analytical part weakly, ran only a few stochastic realizations, and sometimes replaced exact stochastic simulations with solving ODEs. For the more complex Question 5, it did not identify the vaccination threshold and did not model the network in a suitable way, and the parameter choices were not well justified.

Reviewers also noted that some conclusions were misleading as they were not supported by the actual simulation results, and the agent relied on prior knowledge rather than evidence. Even when results were used, they could be incorrect. For example, in targeted vaccination, the agent often claimed that

vaccinating all high-degree nodes is necessary, partly because it generated a sparse network prior to computing thresholds. Overall, the single-agent reports were judged as shallow, with missing figures, limited simulations, and insufficient analysis compared to the multi-agent setup.

The single-agent o3 produced better analytical discussions and more polished writing, but the simulations remained limited, and it showed more drift in execution, for example, not consistently using the provided stochastic engine or not using literature and online search when needed. We found clear inconsistencies in some cases, such as claiming that coexistence was confirmed by stochastic simulations when it was not, or that FastGEMF conducted multiple realizations, whereas the repository's Python scripts indicated a deterministic custom run and even an incorrect model in parts. Similar to GPT-4.1, it still failed to correctly solve the targeted vaccination problem in Question 5, and reviewers concluded that depth and experimental rigor were insufficient for the harder tasks.

It is also necessary to note that, to avoid violating OpenAI's usage policy when using the o3 model, we changed the questions from virus spread to meme spread. Otherwise, the workflow completion rate of single-agent-o3 was less than 30%. We did not consider this API error in our benchmarks for evaluating the o3 workflow completion rate. It is important to emphasize that this change in context does not affect the validity of the comparative results, as epidemic spread can easily be generalized for social cases, such as when a meme spreads over a population.

Overall, human reviews and the task level outcomes show that EpidemIQs addresses the questions more reliably than either single-agent baseline. The single-agent systems were more likely to drift, miss key requirements, and fail in complex scenarios, whereas EpidemIQs maintained greater depth and end-to-end completeness. Table IV reports the computation statistics, showing that these gains are achieved at a cost comparable to single-agent GPT-4.1 and at less than half the cost of single-agent o3, while also improving interpretability by making the reasoning steps explicit.

C. Discussion of Results

Although EpidemIQs demonstrated its applicability and produced high-quality results and manuscripts throughout our evaluation, it is not designed to replace human authorship in scientific research. Its main goal is to serve as an assistant to human experts, facilitating the research process for testing and implementing ideas in epidemic modeling. EpidemIQs can still make mistakes ($\sim 79\%$ task success rate), and AI-generated prose can be highly persuasive, as [52]–[54] discussed, so human oversight remains essential to safeguard accuracy and integrity, and generated papers are meant to provide a comprehensive and unified result of the framework, facilitating the analysis of outcomes.

Across all of our experiments, we found no major instances of hallucination in EpidemIQs outputs, and performance was

TABLE III: Overall Score for AI and Human Expert Evaluation of Generated Results

Model	Question 1		Question 2		Question 3		Question 4		Question 5	
	AI	Human	AI	Human	AI	Human	AI	Human	AI	Human
EpidemIQs	9.0±0.1	8.0	9.0±0.28	8.2	9.2±0.21	7.8	9.0±0.15	8.0	8.96±0.23	8.1
single-agent-GPT4.1	8.4±0.63	5.4	8.4±0.52	5.3	7.9±0.9	4.8	7.3±1.8	4.6	7.9±1.4	5.2
single-agent-o3	8.2±0.63	5.8	8.6±0.55	6.3	8.5±0.2	5.5	8.4±0.36	5.8	8.4±0.45	5.2

TABLE IV: Comparison of Average Computation Statistics Across All Questions Over Three Different Implementations

Method	Time (s)	Tokens (K)	Cost (\$)
EpidemIQs	1190	870	1.57
single-agent-GPT4.1	214	312	0.91
single-agent-o3	436	165	4.13

remarkably consistent. Since LLMs are non-deterministic, they may tackle the same question slightly differently each time; however, no substantial drift or incorrect reasoning was observed. For example, in Question 5, Model Scientist sometimes designs an SIRV (V represents vaccinated state) model, and at other times an SIR model, or Network Scientist may construct networks with different degree distributions yet the same target centrality. These variations are valid and do not diminish performance; they simply reflect the natural diversity of approaches to a single problem. However, there were errors observed during the *Analysis* phase. For example, in Question 3, the Data Expert misinterpreted an average recovery count below one as a fraction rather than a discrete number of individuals. However, the Data Scientist detected this inconsistency and deferred to the Vision Expert, whose interpretation matched both the simulation protocol and analytical expectations. While such checks limit the impact of isolated calculation errors, occasional mistakes by *DataExpert* indicate that automated data analysis may be a part of further improvement.

The framework also demonstrated sound awareness of its computational constraints. For example, in Question 3, we explicitly restricted simulations to the FastGEMF engine. The Simulation Scientist adhered to this limitation, employing FastGEMF exclusively for static networks, and it was explicitly noted in the final report that software limitations prevented temporal network experiments. This transparency is critical, as it prevents the framework from falsely claiming task completion. In contrast, such issues were observed in the single-agent implementation, which incorrectly claimed task accomplishment. Furthermore, the reliability of scientific outcomes often hinges on the algorithms employed, which can be inherently complex. For instance, in one of the trials in Question 3, the Simulation Scientist designed a custom engine to simulate disease spread over a temporal network but erred in updating the state ordering. This led to the premature recovery of initially infected nodes, preventing further spread and resulting in an underestimation of the final outbreak size and probability. Although the Data Scientist noted a significant discrepancy between simulated and expected analytical results, it incorrectly attributed the difference to the effects of temporality rather than the engine's design flaw.

To further investigate the agents' attention to scientific details, in Question 3, we explicitly specified the constant activity rate (and not activity probability per unit time), $\alpha = 3$, which is unlikely in real-world case studies and rare in literature, but was chosen to test the agents' attention to subtle details. However, the Network Scientist incorrectly creates the network edges with the probability of a node being activated as $p = \alpha \Delta t$, with $\Delta t = 1$, as the first order Taylor series approximation of $p = 1 - \exp^{-\alpha \Delta t}$. This approximation is acceptable only if $\alpha \Delta t \ll 1$ (However, if α is probability $\alpha \Delta t$ is exact). Therefore, it was invalid for $\alpha \Delta t = 3 > 1$, resulting in all nodes being activated each time, instead of having exact probability ≈ 0.95 .

These observations emphasize two points:

1) the importance and impact of providing appropriate tools to obtain consistent and reliable results, as system's lowest task success rate was in Question 3, which was not the most advanced but lacked both tools and expert knowledge and 2) the continued need for a human expert to validate results, because agents can still make mistakes that are difficult to detect and may produce persuasive responses that justify incorrect conclusions. In the next section, we categorize the system's failure cases and error modes.

1) *Taxonomy of Failures and Errors*: To rigorously analyze the gaps in the framework, we categorize failures into logic, assumption, algebra, data interpretation, and hallucination.

• **Logic or Methodological Error**: Defined as unjustified logical leaps or invalid mathematical approximations, these errors were the most problematic as they often preserved workflow continuity while compromising scientific validity, and include about 60% of cases where the system failed to achieve the objective correctly. The stress-test study discussed in Question 3 was an example of an invalid approximation, resulting in a probability deviation of 0.05. Such a subtle logical error propagated undetected through the simulation and analysis phases, as the procedure remained syntactically correct. Another example is mapping errors, where severe logical mismatches were often detected. In one temporal network trial, the agent incorrectly mapped the analytical mean degree of a single snapshot ($k \approx 0.4$) to the aggregated static network model. This led to a sparse aggregated network and the isolation of 70% of nodes. Unlike the approximation error, this anomaly was diagnosed during the simulation phase through a reflection loop that prevented it from contaminating the final report. Algorithmic flaws are another form of logical error, such as premature recovery of infected

individuals in the custom simulation engine used to spread over a temporal network. These errors are usually detected and resolved because they lead to unexpected results or execution errors, but they may still propagate silently, as in the aforementioned case, which resulted in a lower epidemic size.

- **Assumption Errors:** These errors involve the introduction of convenient but incorrect premises that simplify the problem but invalidate the specific inquiry, and were about 30% of the task failure cases. For example, in targeted vaccination of Question 5, a failure mode was the assumption that the contact network followed a Poisson degree distribution in the discovery phase. While computationally convenient, this assumption caused the targeted vaccination of degree-10 nodes to be inefficient or impossible due to scarcity. In EpidemIQs, this failure was detected by designing two model setups and testing both the Poisson and a configuration model contact network. During the simulation and analysis phases, it was observed that the Poisson model yielded insufficient degree-10 nodes and correctly identified that the configuration model (with $\approx 11\%$ degree-10 nodes) was required to observe the threshold effect. In contrast, the iterative single-agent baseline consistently adhered to the Poisson assumption, which is a leading reason for the 0% task success rate on this objective.
- **Algebra and Data Interpretation Errors:** While raw algebraic calculation errors were rare, we observed data interpretation errors during the analysis phase, usually due to complexity or large data sizes. For example, in the threshold vaccination question, the Data Expert occasionally misclassified vaccinated individuals as recovered, leading to a larger final size, mainly because in that *SIR* model setup, *R* included both initially vaccinated and recovered nodes. These errors were typically rare and were often corrected during ReAct refinement or cross-validation using Vision Expert.
- **Hallucination:** Defined as confident output that is not true or not supported by evidence, was minimal in *EpidemIQs*' results and strictly limited to time-to-time citation hallucination, generating plausible but non-existent references to support its claims. This stands in sharp contrast to the iterative single-agent baseline, which frequently exhibited process hallucination, particularly regarding task accomplishment when it had actually failed, such as claiming to have run a stochastic simulation when it had only solved a mean-field ODE.

Our classification aligns with recent evaluations of LLM reasoning capabilities [55], which identified logic and assumption errors as the primary obstacles in complex problem-solving tasks of LLMs. Our evaluation reveals that the hierarchical scientist-expert and multi-phase design successfully mitigated failure cases and demonstrated higher failure containment than iterative single-agent, but remains vulnerable to subtle errors, approximations, or assumptions that can propagate across phases.

D. Limitations and Future Directions

To the best of the authors' knowledge, our framework is unique, with no direct comparators. However, a comparison in Question 1 with The AI Scientist, a general tool for scientific discovery from ideation to paper writing [31], showed that it was unable to successfully address the task. It analyzed the ODE-based analytical component but struggled with stochastic simulations over the network, producing hallucinated results and conclusions. The Jupyter Agent 2, as an AI coding agent, fell short in addressing analytical aspects. In contrast, Denario [48] showed better performance on analytical tasks but struggled with stochastic simulations and threshold vaccination in Question 5. It is essential to note that the performance of these AI tools was primarily evaluated in areas other than epidemics, including diffusion models and transformers, coding tasks, and biomedical tasks. This emphasizes that this comparison does not imply overall superiority of one framework over another, but highlights the unique benefits our framework can offer. Summarizing the main limitations of the current work, we have:

- Although no major hallucinations were found in the results, a few hallucinated references did appear in the final report. Moreover, while the reviewers praised the papers' comprehensiveness, they also noted repetitive content that lengthened reading time and, at times, tables that were large, with portions extending beyond the page, suggesting a need to improve the report-generation process.
- The performance of the *EpidemIQs* is highly dependent on the LLM models. While our framework is model-agnostic, as long as these models can support structured outputs and tool calling, the results will vary depending on the LLM used. Since LLMs are word sensitive, a promising future direction is to use the current multi-agent framework as a teacher model for prompt optimization when integrating new LLMs.
- The literature review relies only on the abstract and summary of key findings, rather than delving deeply into the full content of the papers. Future frameworks capable of thoroughly analyzing the full content of each paper could benefit from deeper data retrieval.
- The current framework focuses on epidemic modeling over complex networks as a proof of concept. Extending it to broader methods, such as Agent-Based Models (ABMs), individual-based models, or statistical and data-driven approaches [56], [57], can significantly increase the applicability of the framework. Such extensions can be enabled by equipping the modeling and simulation phases with new tool interfaces and corresponding one-shot examples, allowing the framework to replace FastGEMF with alternative engines. In addition, running multiple modeling pipelines in parallel, where each is based on a different modeling paradigm, would produce an ensemble of outcomes, potentially improving robustness and reducing uncertainty in the final results.
- The agents' performance hinges on the tools at their disposal. As noted in Question 3, with the lowest task success rate of 65%, their performance and reliability decline when they encounter highly technical problems without the requisite

expert knowledge and appropriate tools. This shortcoming explains why the general-purpose module proposed by [31] fails in Question 1.

- AI evaluation in the automated review process shows divergence from human expert reviews, aligning with other works' findings, such as Agent Laboratory [27]. However, in contrast to the near-human performance reported in [31], this suggests that human involvement is necessary. While an LLM-as-Judge can be beneficial for providing high-level insights, it cannot be fully trusted, since it relies on superficial patterns rather than robust analysis criteria. Furthermore, AI evaluation shows a lower tendency toward harsh penalization of incorrect logic or weak experimental rigor than human experts, who usually place greater weight on internal consistency and methodological soundness. Another reason is that AI assessments rely solely on the manuscript text, whereas human reviewers often verify the reported results against the generated scripts and outputs, allowing them to assess whether the findings align with the claimed results.
- While our multi-faceted evaluation system (Section IV) provides a practical approach that could inform future standardization efforts, there is a lack of widely accepted, statistically grounded benchmarks for end-to-end agentic systems, specifically in the epidemic field. Addressing shared tasks, scoring rubrics, and statistically robust evaluation protocols is an important limitation and a direction for future work.
- We did not include ideation in our framework, primarily because epidemic research is highly complex and interdisciplinary. Generating new research ideas requires careful consideration of resources, advanced tools (see Limitation 4), expert knowledge across multiple domains, and access to diverse data sources, such as mobility, weather, and epidemiological data. For example, in vector-borne diseases, mosquito or bird populations play a critical role in epidemiological modeling. To the authors' knowledge, such ideas can quickly become too complex for current agentic AI systems to handle appropriately, given concrete bottlenecks such as the lack of structured access to high-quality mobility/weather/entomological data, difficulty in reasoning across coupled mechanistic models, or access to specific tools. Therefore, we focused on specific network-based epidemic modeling tasks as a proof of concept to showcase the framework's potential benefits.

E. Ethical Concerns and Challenges

Therefore, although EpidemIQs demonstrated the great potential of agentic AI to accelerate epidemic modeling research and allow researchers to focus on high-level ideation, this promise also introduces new ethical challenges that must be taken seriously. Specifically:

- **Scientific Misinformation:** With easy access to advanced modeling tools, the primary risk is the production of incorrect or low-quality papers. As our evaluation demonstrated, EpidemIQs, like any other AI system [34], [48], [49], can produce outputs that appear plausible and methodologically sound in structure and tone but contain errors (Section V-D). For example, as detailed in our failure analysis (Section V-C1),

an algorithmic flaw in one trial led to the premature recovery of infected nodes, resulting in a significant underestimation of the final outbreak size. In a real-world policy setting, overreliance on such a plausible-sounding but technically flawed conclusion could lead to the premature relaxation of public health interventions, thereby increasing overall risk to the population. The other example was when the iterative single-agent incorrectly concluded that herd immunity through high-degree vaccination cannot suppress the disease because of an incorrect assumption about the contact-degree distribution. This highlights the risk of misleading persuasiveness, where the AI's confident tone in its conclusion masks critical underlying mathematical errors that can lead to public misinformation.

Furthermore, the low computational cost of generating a paper for ($\approx$ \$1.57) poses a risk of flooding the scientific record with low-quality or fabricated studies. If unchecked, this could overwhelm the peer-review system, creating a denial-of-service effect on human experts who must verify these outputs. While efforts have been made to detect AI-generated text with high confidence, AI detection remains challenging due to the complexity and continuous evolution of LLMs [58].

- **Dual-Use and Biosecurity Risks:** In the domain of epidemiology, automated modeling tools possess inherent dual-use risks. For example, in the threshold vaccination case, the same reasoning capabilities might, in theory, be inverted by malicious actors to identify optimal conditions for pathogen spread. Although EpidemIQs is currently constrained to standard epidemic models as a proof-of-concept and does not have access to health datasets, future iterations integrated with real-world mobility or health-related data must be gated by strict access controls to prevent the automated optimization of harm. Some guardrails are already built into LLMs to prevent misuse for potentially harmful purposes. However, this trade-off limits the benefits of capable AI models for legitimate, beneficial purposes. A more sustainable solution would be selective, credentialed access to capable models for safety-critical research.
- **Authorship:** The emergence of AI tools introduces new complexities to scientific authorship. While AI may generate content, accountability for scientific validity can not be delegated to the agentic system. Authorship must be shifted from content generation to rigorous verification. EpidemIQs is designed strictly as an AI research assistant subject to human oversight, assuming no authorship rights. Human researchers must explicitly state the use of agentic systems and are responsible for auditing the outputs, verifying assumptions, and validating artifacts and conclusions. Therefore, responsibility and authorship remain with human researchers who must audit and verify the AI system's outputs, including simulations, derivations, and logical reasoning of the AI framework.

To mitigate some of these risks, we have implemented automatic watermarking to prevent authorship confusion in every generated report and a restrictive usage policy that prohibits removing watermarks or submitting EpidemIQs outputs to scientific venues without significant human contribution and verification.

Users must explicitly acknowledge that the system is a research copilot subject to execution errors. A *SecretaryAgent* is implemented as a classifier to trigger EpidemIQs only if the query is safe and within the scope of the framework as an effort to reduce the chance of malicious activity or hallucination.

VI. DATA AVAILABILITY

All the data and results discussed in this paper are provided in Supplementary Sections S1.4 to S1.13 along with reports in Section S4, and the project repository. EpidemIQs is provided as an open-source Python module at github.com/KsuNetse/EpidemIQs. All generated results, configurations of EpidemIQs for five epidemic scenarios discussed here, and five additional cases are available at this repository.

VII. CONCLUSION

In this study, we present EpidemIQs, a multi-LLM agent framework that harnesses the reasoning, planning, and problem-solving abilities of LLMs and integrates them with online and literature retrieval, multimodal data processing, code generation and execution, stochastic simulation, and visual and data analysis. These characteristics, along with its proposed orchestration, enable it to autonomously execute every key stage of the pipeline, from discovery through modeling and simulation to analysis, and finally synthesize the findings into a scientifically structured report summarizing the workflow outcomes. This entire process takes less than half an hour, whereas a human expert would require at least 38.4 hours to complete an execution cycle.

The evaluation of results across questions addressing various aspects of network-based epidemic modeling demonstrates that the framework consistently maintains high performance across our trials. Consequently, EpidemIQs can extend existing analytical approaches to stochastic processes, infer implicit constraints in research questions, and design scenarios that address cases beyond the immediate scope of available tools and expert knowledge.

The comparative analysis of EpidemIQs and the single-agent implementation consistently showed that EpidemIQs outperformed the single-agent variants across all tasks. It achieved a 100% workflow completion rate with an average human-evaluation score of 7.98 at a cost of \$1.57, whereas the single-agent configurations based on GPT-4.1 and o3 obtained scores of 5.06 and 5.68 at costs of \$0.91 and \$4.13, respectively.

At present, EpidemIQs focuses exclusively on epidemic modeling over networks as a proof of concept; nevertheless, it has already demonstrated the potential of deploying LLM agents in a collaborative environment. This opens an exciting path for further performance gains by integrating advanced tools, such as deep learning techniques, probabilistic methods, and data-centric models, heralding a promising future for epidemic modeling research. Moreover, because the framework's accuracy and performance depend heavily on the underlying LLMs, the rapid advances in this field are likely to yield substantial improvements for the entire multi-agent system, as our framework is designed to be model agnostic.

EpidemIQs, while demonstrating strong performance and

producing high-quality output, is not intended to, and should not, replace human authorship in the scientific process, as it still makes mistakes and should not be fully trusted; it is best regarded as a highly capable assistant. Researchers in epidemic modeling can use it to test and implement their ideas quickly and at low cost, freeing them from time-consuming tasks such as setting up simulations or writing repetitive code and allowing them to focus on the conceptual and creative aspects of their work.

Future works should consider the broader applications of epidemic modeling by integrating real-world data and other advanced computational tools into the model design to address open-ended real-world problems and evaluate its applicability for forecasting current outbreaks. Additionally, future work should address ethical concerns regarding the misuse or dual use of such a framework, as automated modeling and report generation could be exploited to produce misleading forecasts or scientifically plausible misinformation, intentionally or unintentionally influencing public perception or policy decisions. The system's ability to simulate outbreak dynamics at scale may also lower barriers for malicious actors to explore harmful scenarios or identify vulnerabilities in public health defenses. Additionally, integration with sensitive epidemiological data poses privacy risks, emphasizing the need for implementing appropriate safeguards.



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